

Pinned group summary: $S0(n=4, q=1)$

Pinning-Sympy automated output

July 6, 2026

Contents

1	Basic attributes	2
1.1	Symmetric bilinear form	2
1.2	Generic Lie algebra element	2
1.3	Generic torus element	2
1.4	Trivial characters	2
2	Root system	3
2.1	Dynkin diagram	3
2.2	Cartan matrix	3
2.3	Roots and coroots	3

1 Basic attributes

Group	$SO(n=4, q=1)$
Matrix size	4×4
Rank	1

1.1 Symmetric bilinear form

Dimension	4
Witt index	1
Primitive element	N/A
Epsilon	N/A

Matrix:
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & c_0 & 0 \\ 0 & 0 & 0 & c_1 \end{bmatrix}$$

1.2 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & 0 & -c_0 x_{21} & -c_1 x_{31} \\ 0 & -x_{00} & -c_0 x_{20} & -c_1 x_{30} \\ x_{20} & x_{21} & 0 & x_{23} \\ x_{30} & x_{31} & -\frac{c_0 x_{23}}{c_1} & 0 \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & \frac{1}{t_0} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

1.4 Trivial characters

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	A1
Reduced	True
Simply laced	True
Number of roots	2

2.1 Dynkin diagram

$$\textcircled{0}$$

2.2 Cartan matrix

$$[2]$$

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
$(-1, 0, 0, 0)$	$(-2, 2, 0, 0)$	No	$-$	No
$(1, 0, 0, 0)$	$(2, -2, 0, 0)$	Yes	$+$	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

No pairs of roots generate positive integer linear combinations.